



Project Forensics Analytics Engine

[Live Demo](#)

[GitHub URL](#)



Forensic Technology Investigation Accelerator



Executive Summary

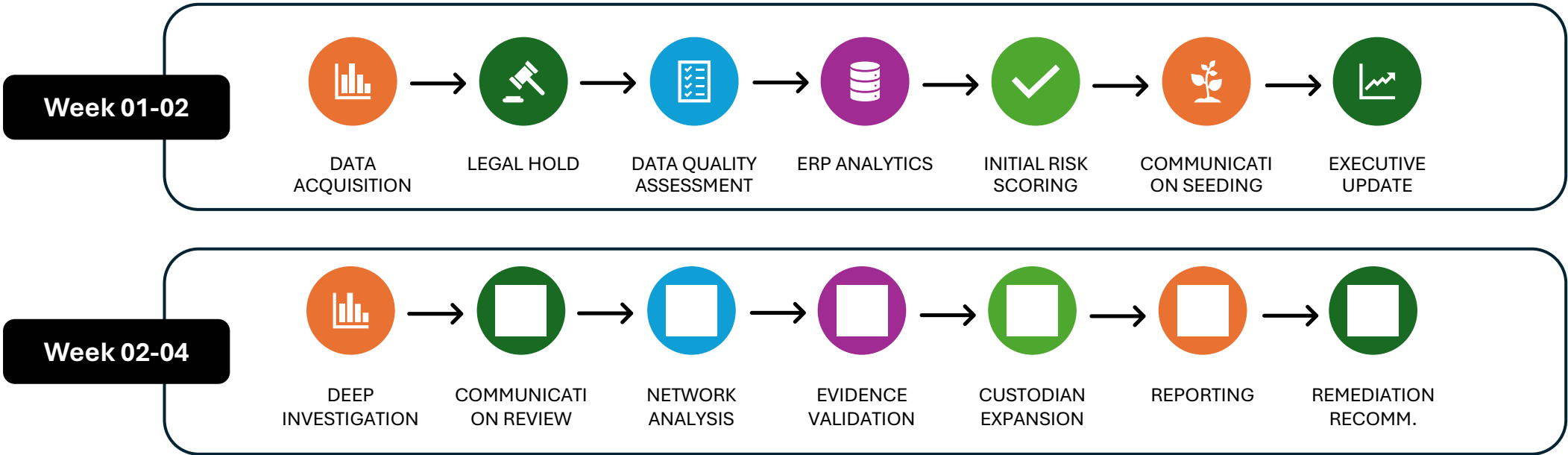
Business Objective

- Deliver a technology-enabled investigation capability that:
- Rapidly identifies high-risk procurement transactions
- Prioritizes investigative effort using risk-based scoring
- Correlates ERP transactions with communications evidence
- Supports Legal, Compliance, Internal Audit, and Executive Management
- Produces regulator-ready evidence with full auditability

Key Outcomes

- Rapid identification of high-risk vendors
- 70–90% reduction in manual transaction review
- Explainable AI-assisted document review
- Defensible investigation methodology
- Executive-ready reporting

Investigation Delivery Approach

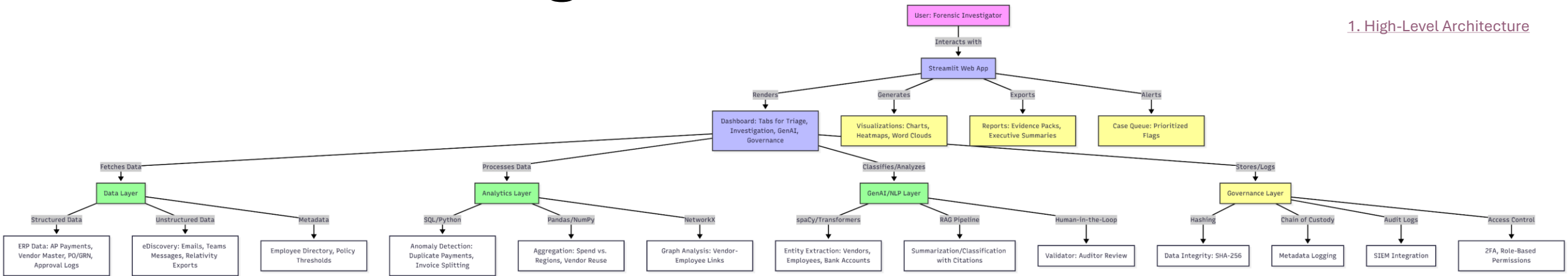


Parallel Workstreams

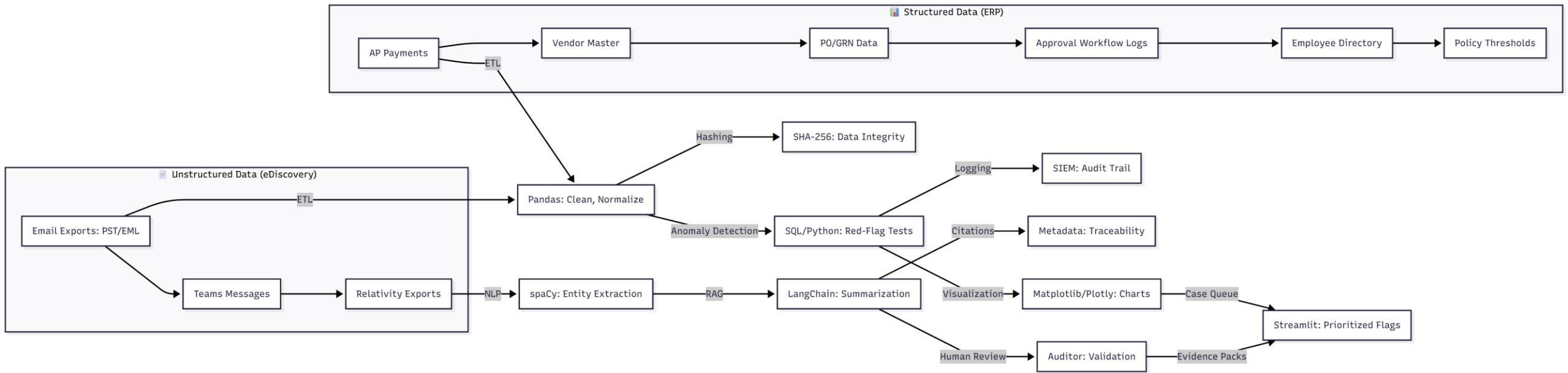
Workstream	Deliverable
ERP Analytics	High-risk transactions
eDiscovery	Relevant communications
AI/NLP	Prioritized document review
Investigation	Evidence validation
Governance	QA, legal defensibility

End-to-End Investigation Architecture

1. High-Level Architecture

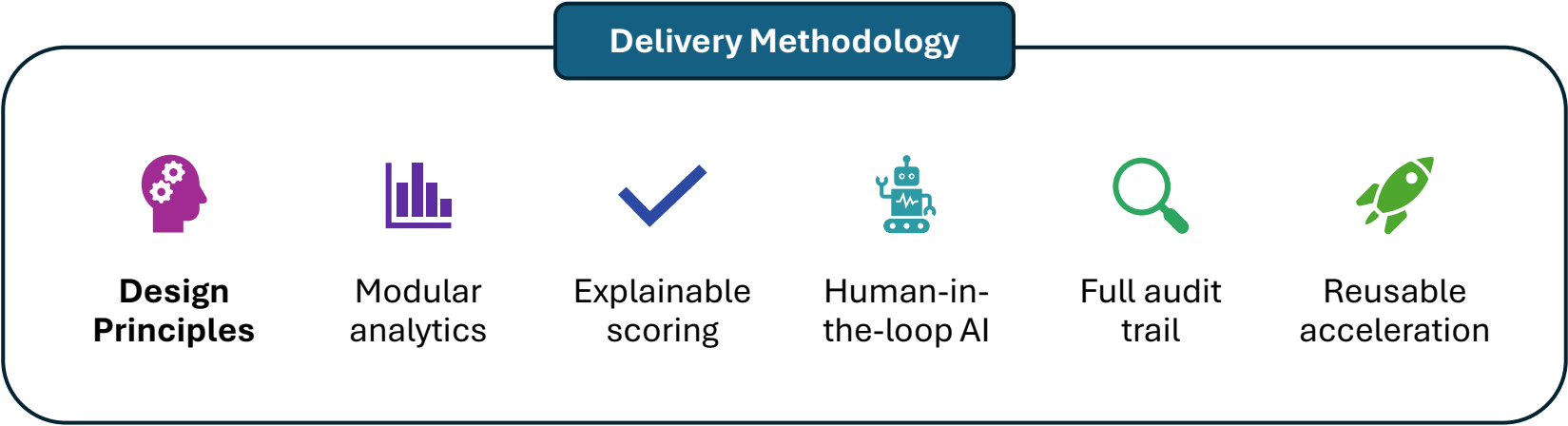
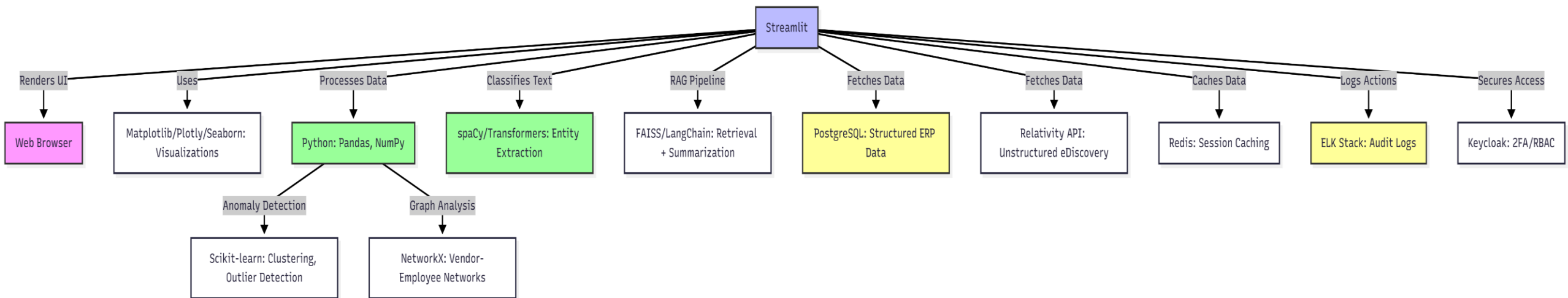


2. Data Pipeline (Structured + Unstruct)



Technology Stack

3. Technology Stack



Risk Analytics & Investigation Logic

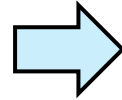
Investigation

Test

Duplicate Payments
Invoice Splitting
Shared Bank Accounts
Vendor Alias Detection
Quarter-End Spend Spike
Approval Override

Business Risk

Duplicate disbursement
Approval circumvention
Shell vendors
Duplicate vendor records
Budget manipulation
Policy bypass



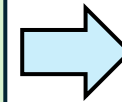
Risk Score

Duplicate Payment (+25)
Shared Bank (+30)
Approval Over (+20)
Vendor Alias (+15)
Communication Indicator (+25)

Total = 115



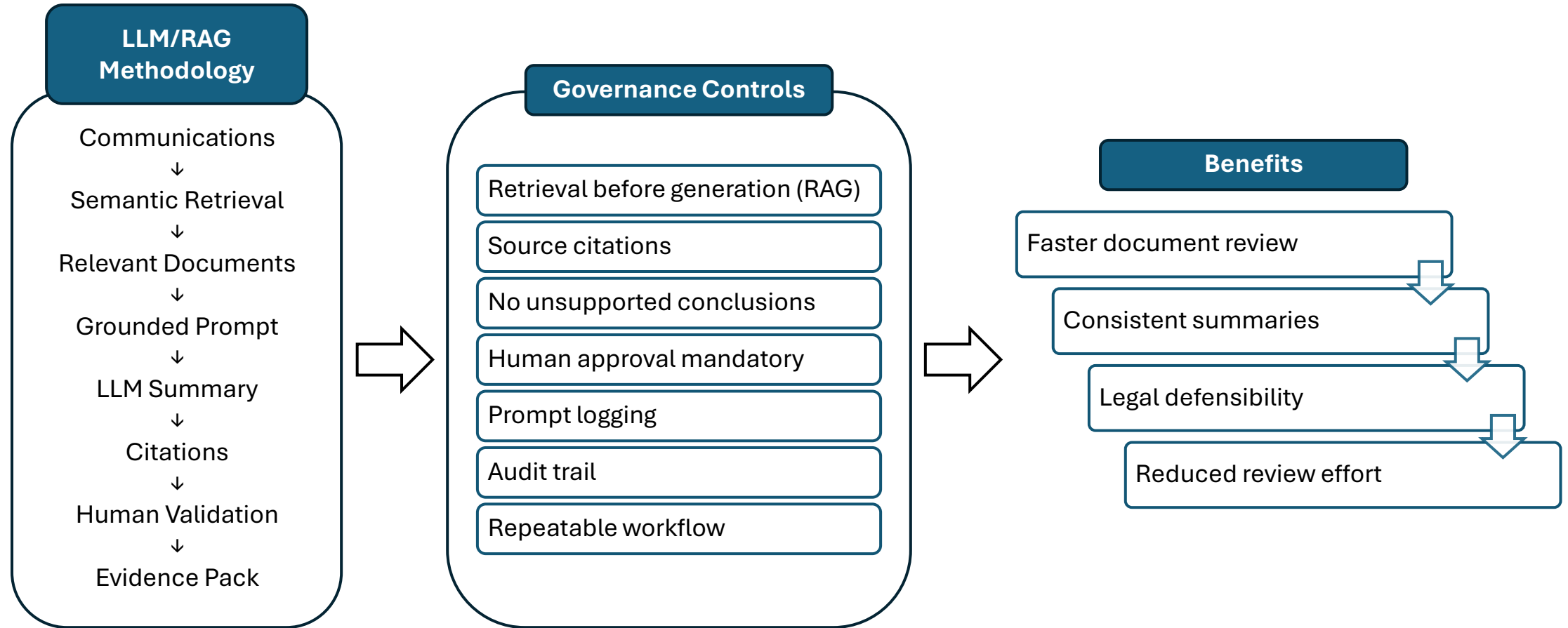
Critical Case



Benefits

- Consistent prioritization
- Reduced false positives
- Explainable outcomes
- Scalable across regions

Defensible GenAI Strategy



Executive Dashboard & Investigation Outcomes

FornsicTech

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[Vendor Intelligence](#)

[Network Graph](#)

[Communication Review](#)

[Case Queue](#)

[Evidence Pack](#)

[Din](#)

Deloitte Forensic Investigation Workbench

Procurement Fraud Investigation Accelerator

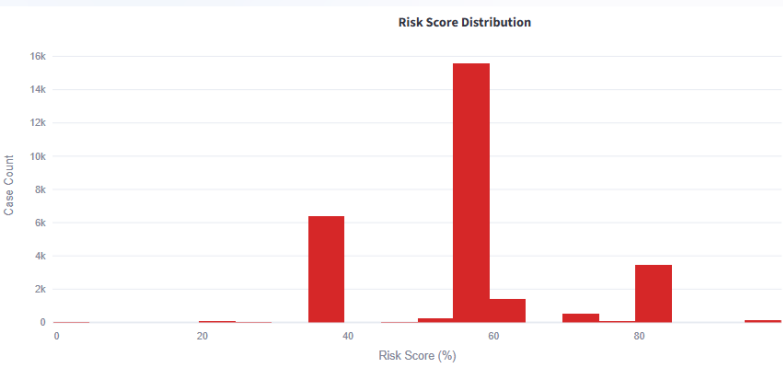
User Login

admin

pass

Manager

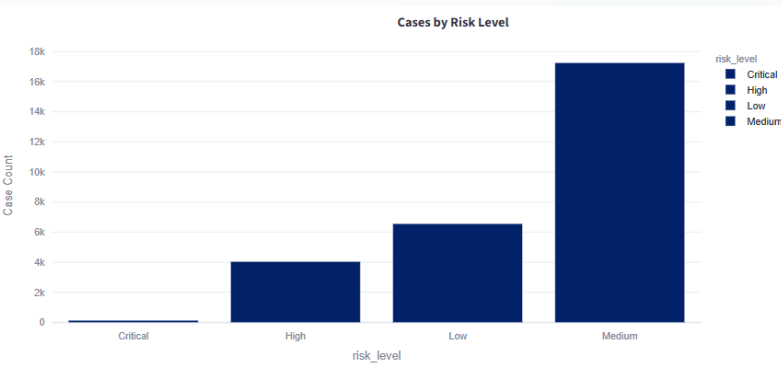
Login



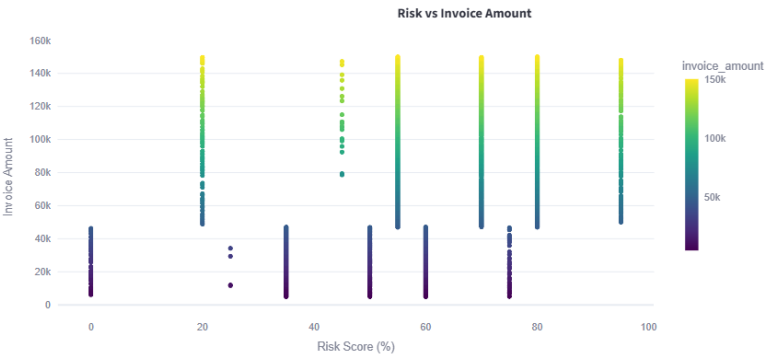
Callout: Critical cases cluster at the upper tail of the score distribution.
Suggested action: Prioritize legal preservation and targeted evidence collection for the highest-scoring queues.



Callout: A small number of invoices materially exceed the typical case pattern.
Suggested action: Validate large-value invoices for split billing or duplicate payment behavior.



Callout: The highest-risk bucket is concentrated in Critical and High cases.
Suggested action: Allocate investigation capacity to the most severe risk bands first.



Callout: Higher-risk cases are associated with larger invoice values and potential control bypass.
Suggested action: Test the largest-value transactions for policy exceptions and unusual approvals.

Executive Dashboard & Investigation Outcomes

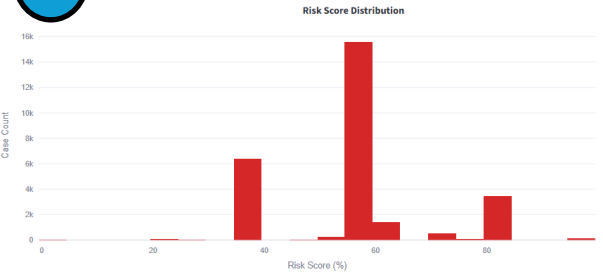
Deloitte Forensic Investigation Workbench

Procurement Fraud Investigation Accelerator

User Login

Manager

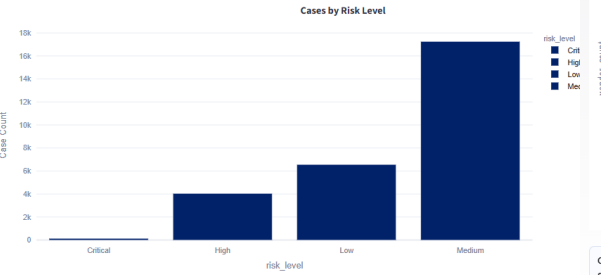
Login



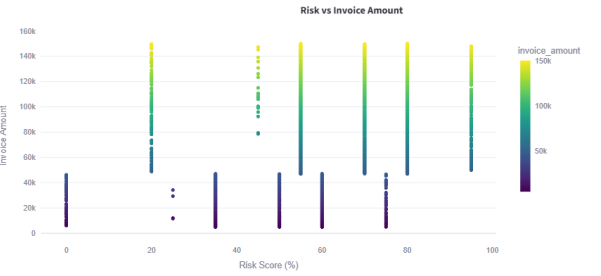
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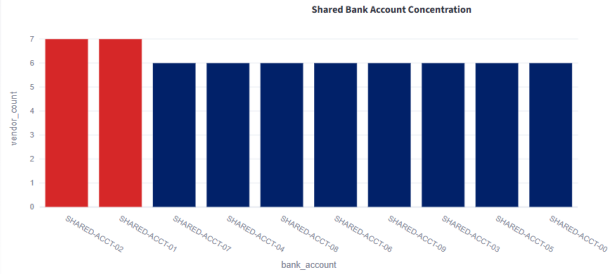
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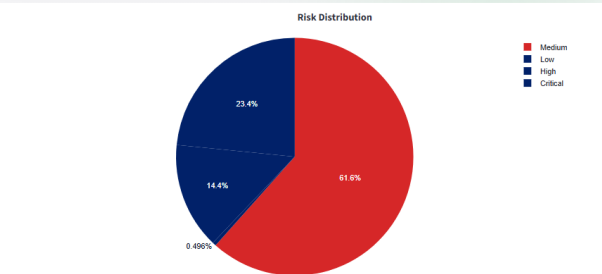
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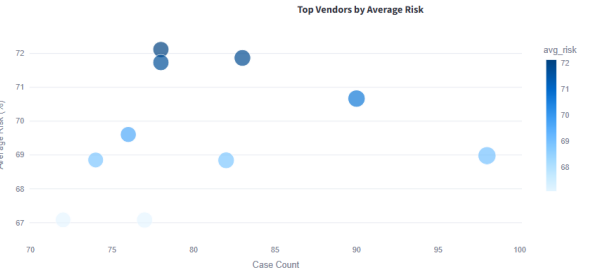
Callout: Duplicate payments and approval overrides are the strongest fraud signals in the current scope.
Suggested action: Focus transaction testing and approval review on the most frequent indicator clusters.



Callout: A small set of bank accounts links multiple vendors and deserves deeper review.
Suggested action: Validate ownership and trace related invoices for the concentrated accounts.



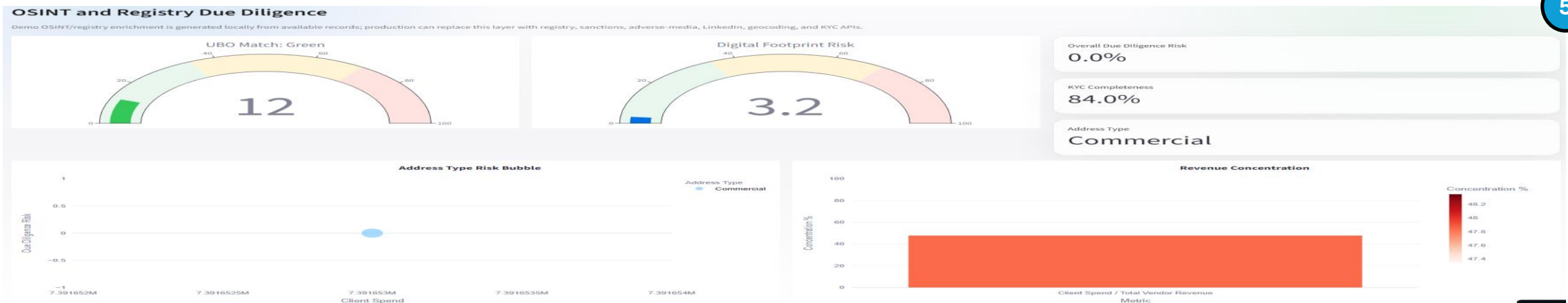
Callout: Critical and High cases represent the majority of the current risk profile.
Suggested action: Direct reserve review capacity toward the highest concentration of elevated-risk items.



Callout: A handful of vendors repeatedly appear in the highest-risk case cluster.
Suggested action: Escalate these vendors for evidence collection and control review.

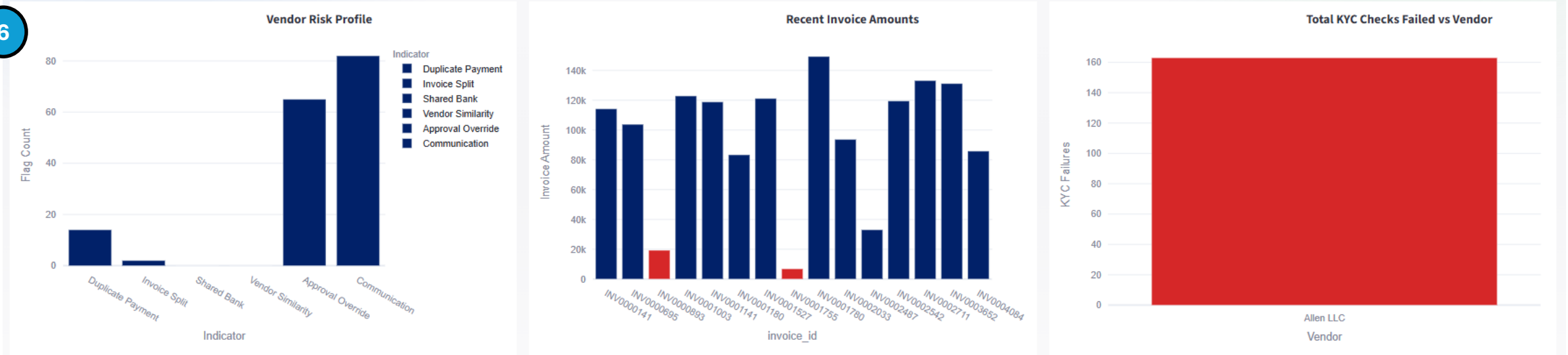
Executive Dashboard & Investigation Outcomes

5



Risk Indicators Spend Profile KYC Checks Failed

6

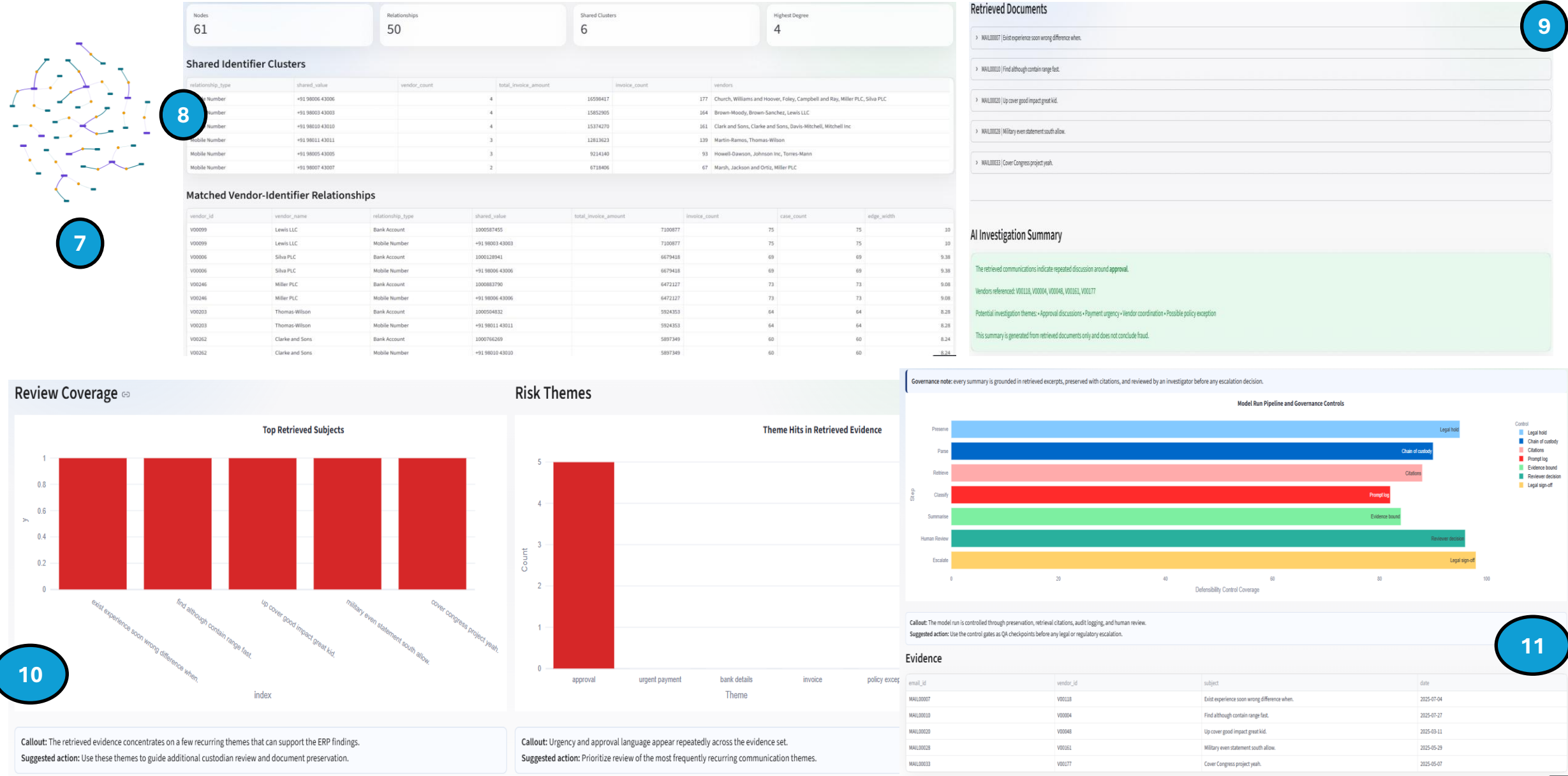


Callout: The vendor shows repeated signal overlap across controls and transaction behavior.
Suggested action: Validate the most frequent indicators with document review and owner interviews.

Callout: A spike in invoice values may indicate threshold manipulation or split billing.
Suggested action: Inspect the highest-value invoices for supporting documentation and approval consistency.

Callout: The derived KYC failure count reflects overlapping red-flag tests that warrant deeper due-diligence review.
Suggested action: Use this as a triage signal for enhanced verification and documentation

Executive Dashboard & Investigation Outcomes



Executive Dashboard & Investigation Outcomes

Case Details

Select Case

CASE00001

Risk Score

95.0%

Priority

P1

Status

Open

Investigation Drivers

☒ Duplicate Payment

☒ Vendor Alias

☒ Communication

☒ Approval Override

Workflow

Assign Investigator

Unassigned

Update Status

Open

Investigator Notes

Save Workflow (Demo)

Evidence Pack Generator

Select Investigation Case

CASE00001

Case Summary

Risk Score

95

Risk Level

Critical

Priority

P1

Investigation Drivers

• Duplicate Payment

• Vendor Alias

• Communication

• Approval Override

Executive Summary

This case has been prioritised based on the composite risk score generated from ERP analytics. Findings require human validation and should not be interpreted as proof of fraud.

Generate Evidence Pack

Din

Ask brief investigation questions grounded in the available ERP, vendor, case, and communications data.

Prompt policy: Din answers from local investigation data and the case background. It keeps answers brief, flags assumptions, and avoids making a final fraud conclusion.

Question

Which vendors has reused mobile numbers and already present as an employees? also add invoice amount, invoice date and how to avoid such cases in future?

Run

Business Value & Future Roadmap

Immediate Benefits

- Accelerated investigations
- Risk-based prioritization
- Cross-functional collaboration
- Standardized forensic methodology
- Improved reporting quality

Next Steps

SAP HANA
integration

Microsoft
Purview
integration

Relativity
APIs

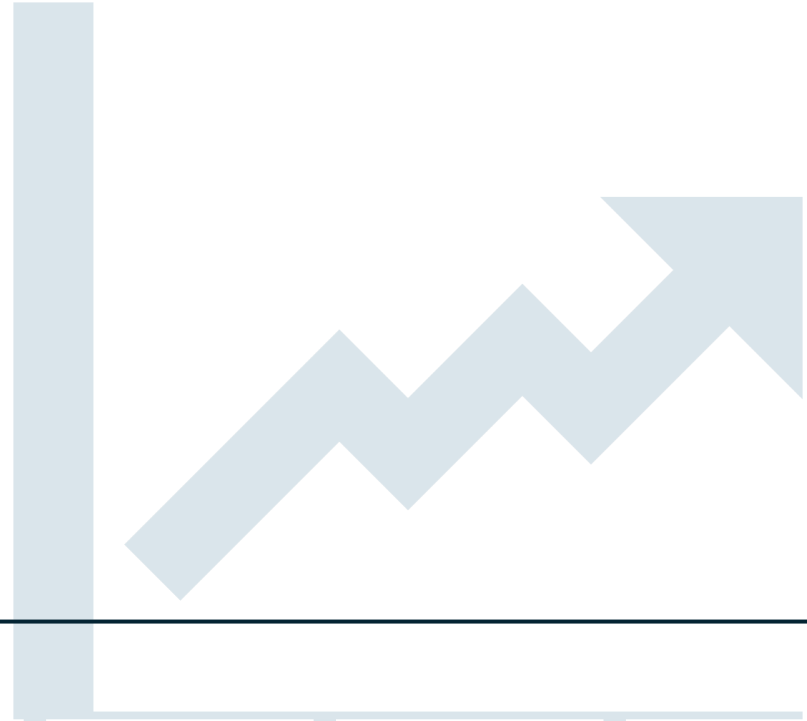
Azure OpenAI

Knowledge
Graph
analytics

Graph-based
anomaly
detection

Continuous
monitoring

Agentic
investigation
workflows



Q/A

Thank You